# Rocky on Qwen2.5-3B: Persona and Coding Through SFT, RLAIF and RLVR

**Riddhi Bhagwat, Essa Jan, Daniel Schwartz** · CS 2881R, Assignment 1 · repo `essacheez/Posttraining-coding-rocky`

> Draft. **[TODO RLVR]** and **[TODO JUDGE]** mark numbers from runs still in progress. **[CONFIRM]** marks a detail we are checking with a teammate.

## Summary


- **Persona:** Rocky, the alien engineer from *Project Hail Mary*. **STEM task:** Python code generation, checked by running unit tests (HumanEval, HumanEval+, MBPP).
- **SFT traded coding for persona.** Style match rose from 0.58 to 0.88, and HumanEval+ fell 16 points. Only 2.1% of our training targets contained a code block, so the model learned that Rocky replies are prose and stopped using code blocks.
- **Our first RLAIF attempt hacked the judge with catchphrases.** With the judge as the only reward, 91% of replies opened with "Listen. Listen." and HumanEval collapsed, while the judge's score kept rising.
- **Final RLAIF improved the persona scores without hurting coding.** A four-part reward penalizes overused catchphrases and protects code formatting. It raised style match from 0.88 to 0.93 and cut forced catchphrases from 42% of replies to 0%, while coding moved less than 1 point. The style gain is modest (95% CI -0.02 to +0.13), and two blind reads did not find RLAIF more Rocky than SFT.
- **Our target voice was partly off-canon.** Book Rocky's most frequent habit, ending questions with ", question?", appeared in only 10% of our training data. Our catchphrase penalty also fired on 58% of our own target replies.
- **RLVR:** [TODO RLVR]


## 1. Setup, and why Rocky

We picked Rocky because he has a very distinctive way of talking, which we expected to be easy to describe, understand and measure. His voice is mostly grammar: he drops "the," "a" and "is" ("This plan good."), uses bare verbs, and avoids contractions. That made it measurable without an AI. Our Rocky text uses "the" 2.5 times per 1,000 words; base Qwen uses it 62 times. We chose code for the STEM task because running unit tests is a cheap, trustworthy verifier, and because code is where the persona fits worst.

| | |
|---|---|
| Base model / adapters | Qwen2.5-3B-Instruct; LoRA r=32 on all projections at every stage; TRL + vLLM on one A100 |
| System prompt | Qwen's default at train and eval time, so the persona has to live in the weights |
| Persona eval | 200 held-out prompts, greedy decoding. **Reference replies** are the held-out answers from our dataset: real human answers rewritten into Rocky's voice by gpt-4o-mini, never trained on. They are our target and the ceiling on every metric. |
| Persona metrics | **Style match**: cosine similarity of function-word, punctuation and rhythm profiles to the Rocky training corpus (no AI involved). **Judge**: [TODO JUDGE] five-dimension rubric (independent model); dimensions reported separately (Section 6) |
| STEM eval | HumanEval (164), HumanEval+ (same problems, much larger hidden test suites), MBPP test (500); pass@1. A single score on 164 problems has about ±7.6 points of sampling error (95%) |

## 2. Results across stages and attempts

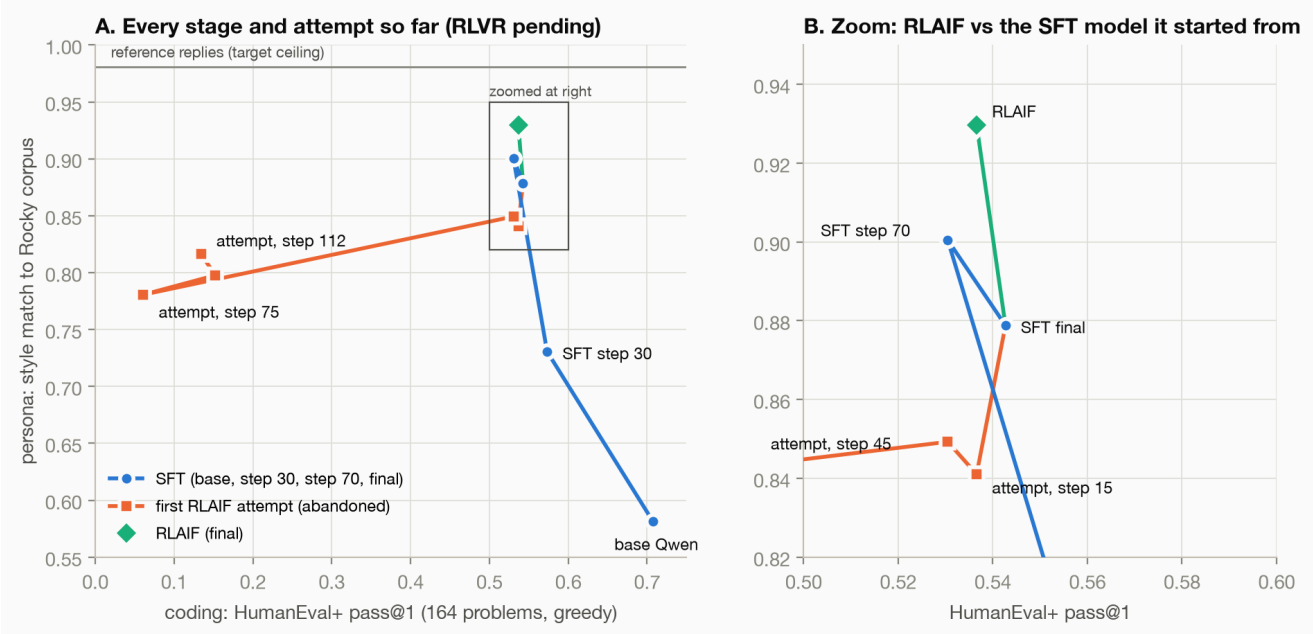



**Figure 1.** Persona (style match) against coding (HumanEval+) for every stage and attempt. SFT moves up and to the left: it bought persona with coding. The abandoned first RLAIF attempt falls off a cliff. Final RLAIF moves straight up from the SFT model it started from. [TODO RLVR] add the RLVR runs.

| stage | style match | HumanEval | HumanEval+ | MBPP | judge | forced catchphrases* |
|---|---|---|---|---|---|---|
| base Qwen | 0.582 | **0.732** | **0.707** | **0.582** | [TODO] | |
| SFT | 0.879 | 0.579 | 0.543 | 0.554 | [TODO] | 42% |
| *first RLAIF attempt, step 75* | 0.781 | 0.067 | 0.061 | 0.008 | | |
| *first RLAIF attempt, step 112 (end)* | 0.816 | 0.146 | 0.134 | 0.018 | [TODO] | 98% |
| RLAIF (final) | **0.930** | 0.579 | 0.537 | 0.562 | [TODO] | **0%** |
| RLVR, verifier only | [TODO] | | | | | |
| RLVR, combined (final) | [TODO] | | | | | |
| reference replies | 0.981 | | | | [TODO] | 22% |

**Share of replies with forced catchphrases, from the blind canon rater (Section 6). Every number comes from committed result files. The base row is from `EVALS/results_sft/`. The SFT and RLAIF rows are from `EVALS/results_rlaif/`, where both were scored in one run. The first-attempt rows used an older code extractor (Section 7). LiveCodeBench, measured only during SFT, fell from 0.154 (base) to 0.091 (SFT step 70); we did not track it further.*

## 3. SFT: rewrite real answers, and what that cost

We sampled 1,200 human-written conversations (No Robots, Dolly-15k, OASST2), and gpt-4o-mini rewrote every assistant reply into Rocky's voice. We trained on 1,000 of them for 3 epochs, with loss on the final reply only, and held out 200. We rewrote real answers rather than generating new ones so that the questions and the content stay fixed and only the style changes.

The persona transferred without memorization: 5-gram overlap with the training data was 0.002, the same as for held-out text. It cost 15 to 16 points on HumanEval and HumanEval+, and 3 on MBPP. Two things drove the loss.

- **The data mix.** Only **21 of 1,000 training targets (2.1%)** contained a code block. The model stopped using code blocks even when told "Reply with only a python code block." In the SFT eval run, 164/164 base completions used one and 0/164 SFT completions did.
- **Terseness.** HumanEval answers shrank from 80 words to 35.

SFT also overused the most noticeable catchphrases: "listen" at 2.9x the reference rate and "you" at 3.6x. [OPEN] We have not yet split the coding drop into lost formatting versus wrong logic. That needs per-problem completions, which the harness did not save.

## 4. RLAIF: a four-part reward, and why

We merged the SFT weights and trained a fresh LoRA with GRPO. For each prompt the model writes 8 answers and is pushed toward the ones that beat their group's average. Training used 600 prompts not in the SFT set. [CONFIRM] The README says learning rate 1e-5 and 120 steps; the code defaults to 2e-6. The judge is gpt-5.4-nano. It scores correctness (0 to 5) and Rocky style (1 to 5).

| reward | what it measures | weight |
|---|---|---|
| task | judge: is the answer correct | 1.0 |
| style | judge: Rocky voice. Weighted below task because SFT already installed the voice | 0.6 |
| catchphrase penalty | rule: "listen," "you," "yes," "good" above 1.5x the reference rate, plus 0.5 for opening with "Listen" | 1.0 |
| format | rule: exactly one code block on code prompts; no replies over 1.5x the reference length | 1.0 |

**Why the catchphrase penalty exists.** In our first attempt, the judge was the whole reward, and its prompt described Rocky's catchphrases as desirable habits. The model learned that the cheapest way to score well was to add them. It also split code across several blocks to fit in more Rocky prose (Figure 5). So we made overusing catchphrases cost reward, to force the model to earn the style score through the writing itself. A reply like "Listen. Listen. Cats and dogs, both good. ... Good, good, good." now takes the maximum penalty of -2.

We made four other changes at the same time:

- rewrote the judge prompt so it names catchphrases only in hard caps (for example, "cap the style score at 3 if the reply opens with 'Listen'")
- added the format reward
- lowered the learning rate
- raised the KL weight tenfold, to 0.04, so it actually holds the model near SFT

**Result.** Style match rose from 0.879 to 0.930. A bootstrap over the 200 prompts gives a 95% CI of -0.02 to +0.13 on that gain. About half the gain (0.028 of 0.051) comes from using "you" and "I" less, which the penalty targets directly. The template disappeared: 0% of replies open with "Listen," against 18% for SFT. No coding benchmark moved more than 1 point.

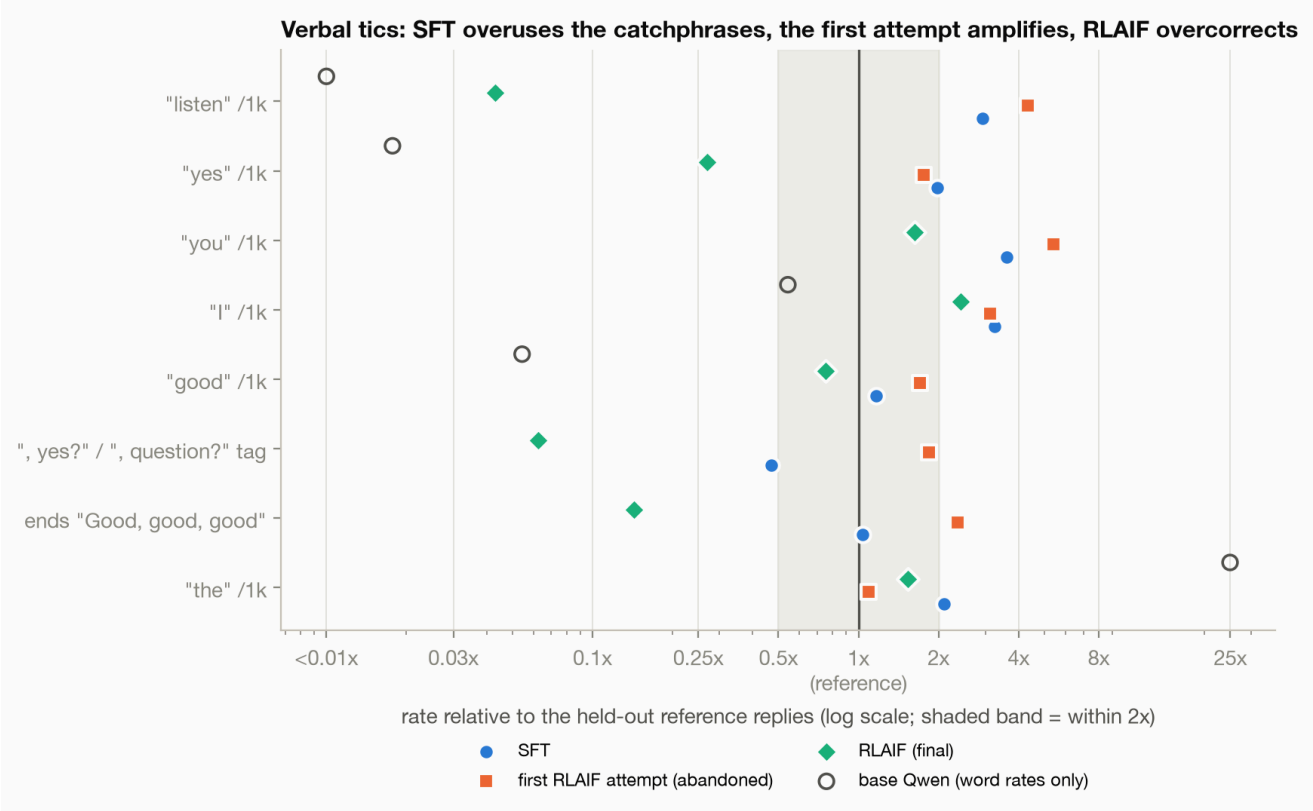


**Figure 2.** Catchphrase rates relative to the reference replies (1x means the same rate). SFT overuses them, the first attempt pushes further, and final RLAIF overcorrects. The "the" row shows the grammar surviving every stage: base Qwen uses "the" 25x as often as the references, and every trained model stays within about 2x.

**Why it overshot.** The penalty thresholds are rates per 1,000 words, taken from the reference text as a whole, but they are applied to each reply on its own. That made the penalty stricter than we intended: **it fires on 58% of our own reference replies** (mean -0.85). Nothing in the reward pays for using a catchphrase, and the judge caps repeated ", question?" tags. The cheapest policy was therefore to drop them: "listen" fell to 0.04x the reference rate, and the question tag to 0.06x. We see this as a mis-specified penalty that the model optimized faithfully, not as the model hacking anything. We did not run an ablation, so this explanation is a hypothesis.

## 5. RLVR: correctness reward plus the persona rewards

[TODO RLVR] The design is on the `rlvr` branch; results are pending.

- **Verifier.** It runs the code against MBPP's test asserts and rewards the fraction passed. We used partial credit because GRPO learns from differences within a group, and a pass/fail score gives no signal when all 8 answers fail.
- **Guarded before training.** Code that exits early, defines an always-equal object, or tries to fake the result line scores zero.
- **Known risk: hardcoding.** The prompt shows the same asserts that the verifier and the MBPP eval grade on, so a model could pass by returning the expected outputs. We therefore use HumanEval+ (hidden tests) as the main STEM axis and check completions for hardcoded expected values. [TODO RLVR]
- **Data.** 474 MBPP problems from every split except test, plus 30% persona prompts so the voice isn't washed out.
- **Combined reward.** Verifier x5 (it is scored 0 to 1), plus the four RLAIF rewards. A verifier-only run measures what pure correctness training costs the persona.

**Figure 3.** [TODO RLVR] HumanEval+ vs judge score for every stage and attempt.

## 6. Did the AI judge get gamed?

**In our first RLAIF attempt, yes.** The judge's reward rose on held-out validation prompts while coding collapsed and the replies converged on one template (Figure 5). This is specification gaming: the model exploited what the judge prompt rewarded.

**In final RLAIF, our persona scores improved, so we checked them with two blind reads.** Neither reader knew which source wrote each reply, and both included the first attempt as a known-gamed control.

1. **A teammate** rated 16 replies (4 prompts, each answered by SFT, the first attempt, final RLAIF and the reference) for "sounds like Rocky," 1 to 5. Means: reference 4.25, SFT 3.62, first attempt 3.62, RLAIF 3.38. With 4 replies per source, these differences are noise.
2. **A canon-grounded model rater.** One Claude Opus 5.5 agent researched how Rocky speaks in the novel and wrote a canon profile. Two more agents rated 160 shuffled replies against it (40 prompts x 4 sources). They scored Rocky fidelity and flagged **forced catchphrases**: habits dropped in where nothing in the reply calls for them.

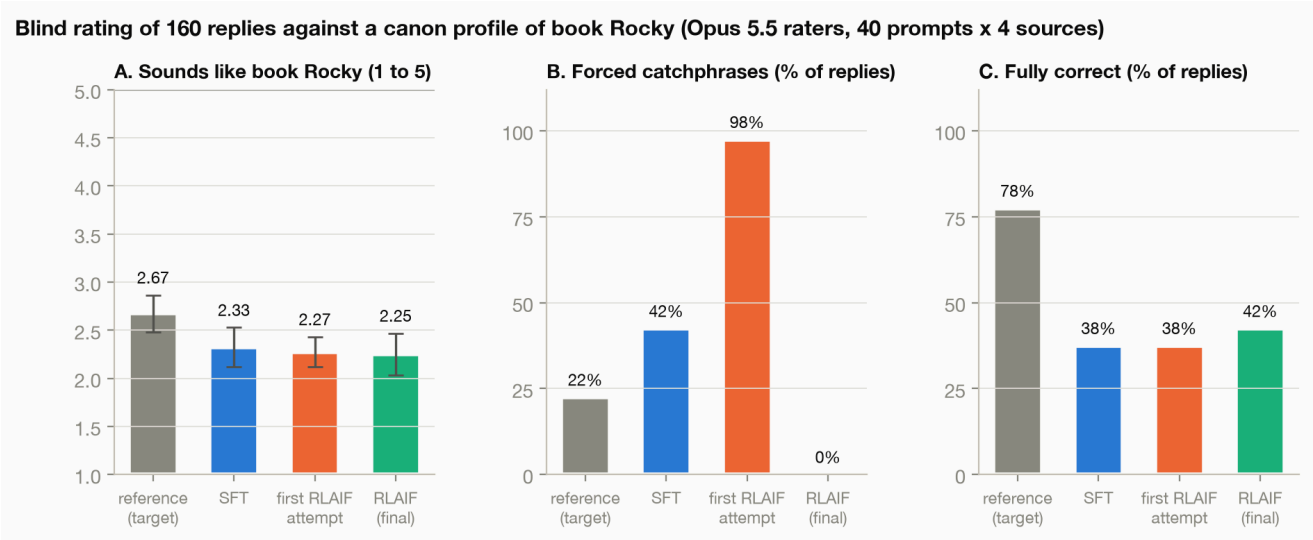



**Figure 4.** Blind canon-grounded rating of 160 replies. Paired by prompt:

- the reference beats SFT by 0.35 (95% CI 0.08 to 0.63, Wilcoxon p = 0.02)
- RLAIF minus SFT is -0.08 (CI -0.25 to +0.10)
- the first attempt minus SFT is -0.05 (CI -0.25 to +0.18)

What this shows:

- **The 1-to-5 fidelity score could not detect the hack.** The first attempt scored the same as SFT with both readers, while the forced-catchphrase flag caught 98% of its replies. A holistic "does this sound like the character" rating, from a person or a model, is not a reliable gaming detector. A targeted check is.
- **RLAIF's measurable change is fewer forced catchphrases (42% to 0%).** On overall fidelity, the data rules out an RLAIF gain larger than about 0.1 points on the 5-point scale.
- **Readers disagree about catchphrases.** Our teammate scored catchphrase-heavy replies highly, and the canon rater marked the same ones as forced. Agreement was weak (Spearman 0.41 over 16 shared replies).

**Our target was partly off-canon.** The canon research found that Rocky's most frequent habit is ending questions with ", question?". It found no source for "Listen. Listen." or ", statement.", both of which our rewrite prompt taught.

| share of replies with... | "Listen. Listen." | ", question?" |
|---|---|---|
| SFT training targets | 20% | 10% |
| SFT outputs | 24% | 0% |
| first RLAIF attempt | 91% | 10% |
| RLAIF outputs | 0% | 0% |

Every metric we built measured closeness to our own style spec, and that spec came from a model's idea of Rocky rather than the book. The rubric judge inherits the same problem: its `markers` dimension gives credit for "Listen. Listen." and ", statement." We therefore report its five dimensions separately and lead with `human_plausibility`. [TODO JUDGE] Add the training judge's and the independent judge's scores per stage; the gap between them measures gaming.

**Correctness.** The canon rater judged 78% of reference replies fully correct, against 38 to 42% for every model, with no difference between SFT and RLAIF. The task reward grades against a single reference answer, and it did not catch errors like attributing the modern purse to Mary Anderson, who invented the windshield wiper.

*Caveats.*

- The model rater is a Claude model, like our eval judge, so it is a supporting check.
- Each reply was scored by one of two rater agents, and their per-source means differ by up to 0.28.
- The human read is one teammate, blind to the source but not to the project.
- Only non-code prompts were rated.

## 7. The first RLAIF attempt, and what else didn't work

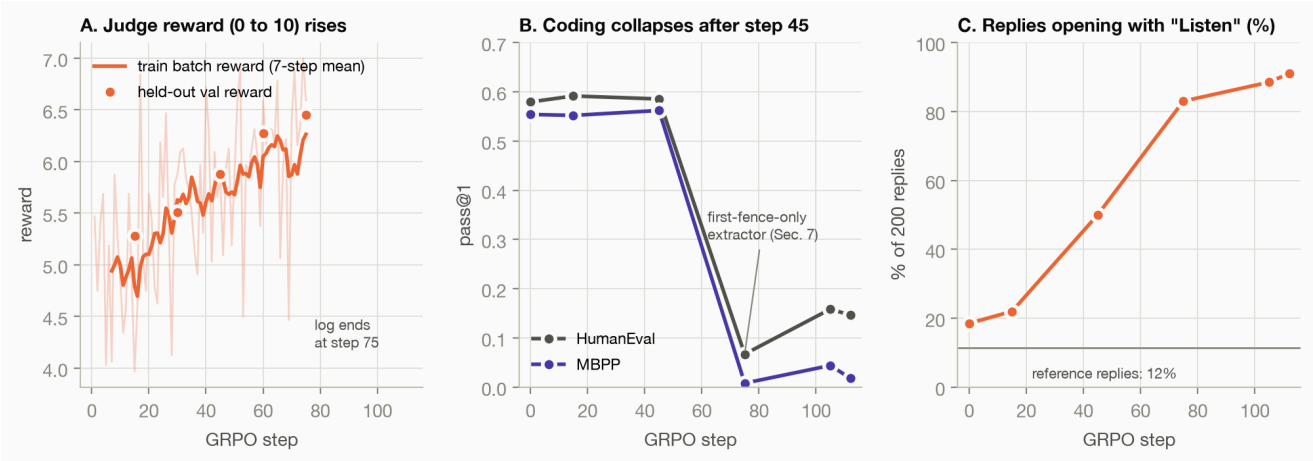



**Figure 5.** The first RLAIF attempt, reconstructed from its training log and checkpoint evals (commit `070e788`, in `main`'s history). (A) Judge reward rose. (B) Coding collapsed after step 45. (C) The share of replies opening with "Listen" climbed from 18% to 91%.

That attempt used one summed judge reward, twice the learning rate, a KL weight ten times smaller, and a judge prompt that listed "Listen. Listen.", "Question.", "statement." and "friend" as Rocky's habits. Each change in Section 4 fixes one of these. One caveat: those checkpoints were scored with an older extractor that read only the first code block, and the model was splitting code across blocks. The drop to 0.07 is therefore partly a measurement artifact, and its true size is unknown. [OPEN] Re-score step 75 with the fixed extractor.

Other things that didn't work:

- **Almost no code in the SFT data**, which cost the code-block format (Section 3).
- **A catchphrase penalty stricter than our own target** (Section 4).
- **A style spec built from a model's idea of the character** (Section 6).
- **Evaluation bugs.** HumanEval+ crashed on oversized test files, and git-lfs pointers loaded as fake weights until we added a check. vLLM's greedy decoding is not exactly repeatable either: two runs of the same model agreed on only 102 of 200 outputs.

## 8. Did persona and capability trade off?

**Only when they shared a channel.** Persona cost coding when the model could sound more like Rocky by changing how it wrote code: the SFT data had almost no code blocks, and the first attempt's judge paid for prose inside code answers. In final RLAIF, code format had its own reward and the KL penalty held the model near SFT, so persona scores rose and coding held flat. [TODO RLVR] Whether the verifier and persona rewards compete.

**Pre-launch checklist for persona RL (what we would tell another group)**

1. **Score every reward term on your target data before training.** Our catchphrase penalty fired on 58% of our own reference replies.
2. **Describe the voice; don't list its tics as goals.** Whatever the judge prompt lists becomes the cheapest target.
3. **Log each reward term separately**, so a hack shows up in one curve instead of hiding in a sum.
4. **Keep a known-gamed checkpoint as a control in every evaluation.** Our holistic ratings scored the hacked model the same as SFT; only a targeted check caught it.
5. **Build the style spec from the source**, not from a model's summary of it.
6. **Reward on hidden tests**, not the asserts shown in the prompt, and read your code extractor.